
History of Artificial Intelligence

Introduction

The **history of artificial intelligence (AI)** is a journey through human attempts to replicate and augment intelligence using machines. This chapter outlines the chronological development of AI from ancient myths to modern machine learning models. You will learn about key milestones, figures, and conceptual shifts that have shaped the field.

A simplified timeline of major events in AI history

Ancient and Classical Roots

Even before machines existed, people imagined artificial beings:

- **Mythological Origins:** In Greek mythology:
 - **Talos**, a bronze automaton, guarded Crete.
 - **Pygmalion's statue**, Galatea, came to life through divine intervention.
- **Philosophical Thought:**
 - **Aristotle** developed syllogistic logic.
 - **Ramon Llull** (1232–1316) designed logical machines to combine concepts mechanically.

Depiction of Talos, the ancient bronze automaton

17th–19th Century: Mechanical Reasoning

Early efforts to build thinking or reasoning machines:

- **Leibniz** imagined a “**calculus ratiocinator**” for reasoning.
- **Pascal's calculator** and **Babbage's Difference Engine** laid foundations for computational machines.
- **Vaucanson's automata** mimicked life in mechanical form.

Babbage's Difference Engine, a precursor to modern computers

Early 20th Century: Logic and Computation

The theoretical underpinnings of AI emerged:

- **Boolean logic** enabled formal reasoning.
- **Alan Turing** introduced the **Turing Machine**, a model of computation.
- In 1950, Turing proposed the **Turing Test** as a measure of machine intelligence.

Turing Machine concept – a foundational model for AI and computing

1950s: Birth of AI as a Field

AI became an organized field of study:

- **Dartmouth Conference (1956)** brought together pioneers like:
 - **John McCarthy** (who coined “AI”)
 - **Marvin Minsky**
 - **Claude Shannon**
 - **Allen Newell** and **Herbert Simon**
- Notable early programs:
 - **Logic Theorist** (1956): First AI program.
 - **GPS**: Aimed to solve general problems symbolically.

Participants of the 1956 Dartmouth Conference – the birth of AI

1960s–1970s: Symbolic AI and Expert Systems

- **Symbolic AI** modeled intelligence using rules and symbols.
- **ELIZA** (1966) mimicked conversation using pattern matching.
- **SHRDLU** (1970) understood commands in a simulated block world.
- **Expert Systems**:
 - **MYCIN** diagnosed bacterial infections using logic rules.

ELIZA program simulating psychotherapy with rule-based dialogue

1980s: Knowledge Engineering and AI Winter

- **Knowledge Engineering** peaked, but systems couldn’t scale.
- **AI Winter**: Overpromises led to funding cuts and skepticism.

Diagram of a typical expert system structure (knowledge base + inference engine)

1990s: Statistical AI and Machine Learning

A new wave emerged based on **data and probability**:

- **Machine learning** replaced rules with algorithms that learned from data.
- **Decision trees**, **Bayesian networks**, and **support vector machines** rose to prominence.
- In 1997, **IBM’s Deep Blue** beat Garry Kasparov in chess.

Garry Kasparov playing against Deep Blue, 1997

2000s: Data, Web, and Robotics

- **Internet and sensors** provided massive datasets.
- **Speech, vision, and robotics** advanced.
- **AI began powering daily tools:** spam filters, search engines, recommendation systems.

How AI started supporting web services in the 2000s

2010s: Deep Learning and Big AI

- **Deep learning** dominated AI research:
 - Convolutional Neural Networks (CNNs) for vision
 - Recurrent Neural Networks (RNNs) for language
- **AlphaGo** (2016) defeated a Go world champion, a major symbolic victory.

Typical architecture of a deep neural network

2020s: Foundation Models and Generative AI

- **Transformers** enabled large-scale language models:
 - **GPT, BERT, Claude, Gemini, LLaMA**
- **Multimodal models:** Generate images, text, video, code.
- Ethical challenges prompted research into **AI safety** and **alignment**.

Transformer architecture used in modern foundation models

Summary

The history of AI spans myth, logic, computation, and deep learning. AI has evolved from symbolic systems to massive data-driven models shaping modern life. Recognizing this history helps us understand not only what AI is today, but how it reflects centuries of human ambition and ingenuity.

Further Reading

- Wikipedia: Artificial Intelligence
- Wikipedia: Turing Test
- Wikipedia: History of Artificial Intelligence
- Machine Learning
- Neural Networks
- Wikibooks: Artificial Intelligence
- Wikibooks: Introduction to Machine Learning
- Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach*.